3-Tier Application Deployment on Linux VMs

AIM:-

The aim of this project is to design and deploy a scalable **3-tier web application architecture** by separating the frontend, backend, and database into independent servers. This approach improves security, performance, and maintainability.

Architecture Overview

- **Web Server (Frontend)** → Flask UI (Port 5000)
- **App Server (Backend)** → Flask API + SQLAlchemy + JWT (Port 5001)
- **DB Server (Database)** → MySQL (Port 3306)

STEP 1: Database Server Setup (MySQL)

Install MySQL

sudo su

Switch to root user for administrative tasks

apt update

Refresh package list

```
gowthamazure@database-server:~$ sudo su
root@database-server:/home/gowthamazure# apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1846 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
```

apt install mysql-server -y

Install MySQL server

```
Reading state information... Done
47 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@database-server:/home/gowthamazure# apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
```

Secure MySQL

systemctl enable mysql

Enable MySQL to start on boot

systemctl start mysql

Start MySQL service

```
root@database-server:/home/gowthamazure# systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
root@database-server:/home/gowthamazure# systemctl start mysql
```

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mysql_secure_installation

Apply security settings

VALIDATE PASSWORD COMPONENT can be used to test passwords and improve security. It checks the strength of password and allows the users to set only those passwords which are secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: n

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y

Remove test database and access to it? (Press y|Y for Yes, any other key for No) : y

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y

If it doesn't ask for a root password — that's fine (Ubuntu uses auth_socket authentication).

```
root@database-server:/home/gowthamazure# mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW    Length >= 8
MEDIUM Length >= 8, numeric, mixed case, and special characters
STRONG Length >= 8, numeric, mixed case, special characters and dictionary                  file

Please enter 0 = LOW, 1 = MEDIUM and 2 = STRONG: 0

Skipping password set for root as authentication with auth_socket is used by default.
If you would like to use password authentication instead, this can be done with the "ALTER_USER" command.
See https://dev.mysql.com/doc/refman/8.0/en/alter-user.html#alter-user-password-management for more information.

By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
```

Login to MySQL

Mysql

Open MySQL shell

Create Database & User

CREATE DATABASE appdb;

Create application database

CREATE USER 'appuser'@'%' IDENTIFIED BY 'Password12345';

Create user accessible from any host

GRANT ALL PRIVILEGES ON appdb.* TO 'appuser'@'%';

Give full access to the database

FLUSH PRIVILEGES;

Apply permission changes

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```
root@database-server:/home/gowthamazure# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE appdb;
Query OK, 1 row affected (0.03 sec)

mysql> CREATE USER 'appuser'@'%' IDENTIFIED WITH mysql_native_password BY 'Password12345';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT ALL PRIVILEGES ON appdb.* TO 'appuser'@'%';
Query OK, 0 rows affected (0.02 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)

mysql> EXIT;
Bye
```

Enable Remote Access

vim /etc/mysql/mysql.conf.d/mysqld.cnf

Edit MySQL config

Change:

Find this line:

bind-address = 127.0.0.1

Change it to:

bind-address = 0.0.0.0

```
Bye
root@database-server:/home/gowthamazure# vi /etc/mysql/mysql.conf.d/mysqld.cnf
root@database-server:/home/gowthamazure# systemctl restart mysql
root@database-server:/home/gowthamazure# mysql
```

```
# datadir       = /var/lib/mysql


# If MySQL is running as a replication slave, this should be
# changed. Ref https://dev.mysql.com/doc/refman/8.0/en/server-system-variables.html#sysvar_tmpdir
# tmpdir                = /tmp
#
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address            = 0.0.0.0
mysqlx-bind-address     = 127.0.0.1
#
# * Fine Tuning
#
key_buffer_size         = 16M
# max_allowed_packet    = 64M
# thread_stack          = 256K

# thread_cache_size       = -1
```

Then restart MySQL:

sudo systemctl restart mysql

```
root@database-server:/home/gowthamazure# vi /etc/mysql/mysql.conf.d/mysqld.cnf
root@database-server:/home/gowthamazure# systemctl restart mysql
root@database-server:/home/gowthamazure# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

Verify user and plugin

sudo mysql

SELECT user, host, plugin FROM mysql.user;

You should see:

| appuser | % | mysql_native_password |

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That means Flask's mysqlclient driver can connect successfully.

```
root@database-server:/home/gowthamazure# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SELECT user, host, plugin FROM mysql.user;
+------------------+-----------+-----------------------+
| user             | host      | plugin                |
+------------------+-----------+-----------------------+
| appuser          | %         | mysql_native_password |
| debian-sys-maint | localhost | caching_sha2_password |
| mysql.infoschema | localhost | caching_sha2_password |
| mysql.session    | localhost | caching_sha2_password |
| mysql.sys        | localhost | caching_sha2_password |
| root             | localhost | auth_socket           |
+------------------+-----------+-----------------------+
6 rows in set (0.00 sec)

mysql>
```

STEP 2: App Server Setup (Flask Backend)

Install Dependencies

sudo apt update

Update packages

```
gowthamazure@application-server:~$ sudo su
root@application-server:/home/gowthamazure# apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1846 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1665 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2837 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [656 kB]
```

Install Python and MySQL build dependencies

sudo apt install python3-full python3-pip pkg-config python3-dev default-libmysqlclient-dev build-essential -y

```
root@application-server:/home/gowthamazure# sudo apt install python3-full python3-pip pkg-config python3-dev default-libmysqlclient-dev build-essential -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  2to3 binutils binutils-common binutils-x86-64-linux-gnu blt bzip2 cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev fakeroot fontconfig-config
  fonts-dejavu-core fonts-dejavu-mono fonts-mathjax g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
  gcc-x86-64-linux-gnu idle idle-python3.12 javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libaom3 libasan8 libatomic1
  libbinutils libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0 libctf0 libde265-0 libdeflate0 libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libfontconfig1 libgcc-13-dev libgd3 libgomp1 libgprofng0 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1
  libhwasan0 libisl23 libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-mathjax libjs-sphinxdoc libjs-underscore liblerc4 liblsan0 libmpc3 libmysqlclient-dev
  libmysqlclient21 libpkgconf3 libpython3-dev libpython3.12-dev libpython3.12-testsuite libquadmath0 libsframe1 libsharpyuv0 libssl-dev libstdc++-13-dev libtiff6
  libtk8.6 libtsan2 libubsan1 libwebp7 libxft2 libxpm4 libxrender1 libxss1 libzstd-dev linux-cloud-tools-common linux-libc-dev linux-tools-common lto-disabled-list make
  manpages-dev mysql-common net-tools pkgconf pkgconf-bin python3-doc python3-examples python3-lib2to3 python3-pip-whl python3-setuptools-whl python3-tk python3-venv
  python3-wheel python3.12-dev python3.12-doc python3.12-examples python3.12-full python3.12-venv rpcsvc-proto tk8.6-blt2.5 x11-common zlib1g-dev
Suggested packages:
  binutils-doc gprofng-gui blt-demo bzip2-doc cpp-doc gcc-13-locales cpp-13-doc debian-keyring g++-multilib g++-13-multilib gcc-13-doc gcc-multilib autoconf automake
  libtool flex bison gdb gcc-doc gcc-13-multilib gdb-x86-64-linux-gnu apache2 | lighttpd | httpd glibc-doc bzr libgd-tools libheif-plugin-x265 libheif-plugin-ffmpegdec
  libheif-plugin-jpegdec libheif-plugin-jpegenc libheif-plugin-j2kdec libheif-plugin-j2kenc libheif-plugin-rav1e libheif-plugin-svtenc fonts-mathjax-extras fonts-stix
  libjs-mathjax-doc libssl-doc libstdc++-13-doc tk8.6 make-doc tix python3-tk-dbg
The following NEW packages will be installed:
  2to3 binutils binutils-common binutils-x86-64-linux-gnu blt build-essential bzip2 cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu default-libmysqlclient-dev
  dpkg-dev fakeroot fontconfig-config fonts-dejavu-core fonts-dejavu-mono fonts-mathjax g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base
  gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu idle idle-python3.12 javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libaom3
  libasan8 libatomic1 libbinutils libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0 libctf0 libde265-0 libdeflate0 libdpkg-perl libexpat1-dev
  libfakeroot libfile-fcntllock-perl libfontconfig1 libgcc-13-dev libgd3 libgomp1 libgprofng0 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265
  libheif1 libhwasan0 libisl23 libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-mathjax libjs-sphinxdoc libjs-underscore liblerc4 liblsan0 libmpc3
  libmysqlclient-dev libmysqlclient21 libpkgconf3 libpython3-dev libpython3.12-dev libpython3.12-testsuite libquadmath0 libsframe1 libsharpyuv0 libssl-dev
```

Clone repository

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git clone https://github.com/gowthamsaikadali/vcube-3tier-application-python.git

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@application-server:/home/gowthamazure# git clone https://github.com/gowthamsaikadali/3-tier-application-in-linux-vm.git
Cloning into '3-tier-application-in-linux-vm'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 18 (delta 1), reused 18 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 9.74 KiB | 9.74 MiB/s, done.
Resolving deltas: 100% (1/1), done.
root@application-server:/home/gowthamazure# ls
3-tier-application-in-linux-vm
root@application-server:/home/gowthamazure# cd 3-tier-application-in-linux-vm/
```

Navigate to backend directory

cd app_server

```
root@application-server:/home/gowthamazure# ls
3-tier-application-in-linux-vm
root@application-server:/home/gowthamazure# cd 3-tier-application-in-linux-vm/
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm# ls
app_server  web_server
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm# cd app_server/
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# ls
app.py  create_db.py  create_tables.py  instance  models.py  requirements.txt
```

Setup Virtual Environment

Create isolated Python environment

python3 -m venv venv

```
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# python3 -m venv venv
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# source venv/bin/activate
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# pip install -r requirements.txt
```

Activate virtual environment

source venv/bin/activate

```
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# python3 -m venv venv
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# source venv/bin/activate
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# pip install -r requirements.txt
```

Install Requirements

Install project dependencies

pip install -r requirements.txt

```
root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# source venv/bin/activate
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# pip install -r requirements.txt
Collecting Flask==2.2.5 (from -r requirements.txt (line 1))
  Downloading Flask-2.2.5-py3-none-any.whl.metadata (3.9 kB)
Collecting Flask-SQLAlchemy==3.0.3 (from -r requirements.txt (line 2))
  Downloading Flask_SQLAlchemy-3.0.3-py3-none-any.whl.metadata (3.4 kB)
Collecting PyJWT==2.8.0 (from -r requirements.txt (line 3))
  Downloading PyJWT-2.8.0-py3-none-any.whl.metadata (4.2 kB)
Collecting Werkzeug==2.2.3 (from -r requirements.txt (line 4))
  Downloading Werkzeug-2.2.3-py3-none-any.whl.metadata (4.4 kB)
Collecting PyMySQL==1.0.3 (from -r requirements.txt (line 5))
  Downloading PyMySQL-1.0.3-py3-none-any.whl.metadata (4.4 kB)
Collecting Jinja2>=3.0 (from Flask==2.2.5->-r requirements.txt (line 1))
```

Install MySQL driver for Python

pip install mysqlclient

```
3.2 itsdangerous-2.2.0 typing-extensions-4.15.0
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# pip install mysqlclient
Collecting mysqlclient
  Downloading mysqlclient-2.2.8.tar.gz (92 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 92.3/92.3 kB 2.3 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: mysqlclient
  Building wheel for mysqlclient (pyproject.toml) ... done
  Created wheel for mysqlclient: filename=mysqlclient-2.2.8-cp312-cp312-linux_x86_64.whl size=129532 sha256=5d1d058df42b05e246ab1ac8e2e4857965e6d4de854ddab385970a
b4
  Stored in directory: /root/.cache/pip/wheels/b0/62/00/46e65ad9151c1266f0f03ea020a5a8ea1cbc6e4ebf3329e276
Successfully built mysqlclient
Installing collected packages: mysqlclient
Successfully installed mysqlclient-2.2.8
```

To verify installation:

pip show mysqlclient

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```
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# pip show mysqlclient
Name: mysqlclient
Version: 2.2.8
Summary: Python interface to MySQL
Home-page:
Author:
Author-email: Inada Naoki <songofacandy@gmail.com>
License:
Location: /home/gowthamazure/3-tier-application-in-linux-vm/app_server/venv/lib/python3.12/site-packages
Requires:
Required-by:
```

Configure Environment Variables

export DATABASE_URL="mysql://appuser:Password12345@<DB_PRIVATE_IP>/appdb"

Set database connection string

export SECRET_KEY="supersecretkey"

Set app security key

export APP_HOST="0.0.0.0"
export APP_PORT=5001

Configure backend server host & port

```
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# export DATABASE_URL="mysql://appuser:Password12345@10.0.0.4/appdb"
export SECRET_KEY="supersecretkey"
export APP_HOST="0.0.0.0"
export APP_PORT=5001
```

Initialize Database Tables

python3

open python shell

from app import app, db
with app.app_context():
    db.create_all()

Create required tables in MySQL

```
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# python3
Python 3.12.3 (main, Mar  3 2026, 12:15:18) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from app import app, db
>>> with app.app_context():
...     db.create_all()
...
>>> exit()
```

double click on enter after pasting

To exit

exit()

If no error, tables are created in appdb.

You can verify on DB server:

Mysql

Install mysql-server -y

```
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl libbusiness-isbn-perl libregexp-ipv6-perl libwww-perl mailx tinyca
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64
  libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-server mysql-server-8.0 mysql-server-core-8.0
```

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You can verify on DB server:

USE appdb;

SHOW TABLES; or SHOW DATABASES;

Should show:

| user |

```
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# mysql -h 10.0.0.4 -u appuser -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE appdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+----------------+
| Tables_in_appdb |
+----------------+
| user           |
+----------------+
```

Run app in background (APP SERVER)

nohup python3 app.py > app_server.log 2>&1 &

```
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# nohup python3 app.py > app_server.log 2>&1 &
[3] 5111
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server#
```

STEP 3: Web Server Setup (Frontend)

Install Python

apt update

```
Last login: Sun Mar 29 17:36:33 2026 from 110.225.171.235
gowthamazure@web-server:~$ sudo su
root@web-server:/home/gowthamazure# sudo su
root@web-server:/home/gowthamazure# apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
```

Install Python

apt install python3-full python3-pip -y

```
root@web-server:/home/gowthamazure# apt install python3-full python3-pip -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  2to3 binutils binutils-common binutils-x86-64-linux-gnu blt build-essential bzip2 cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev fakeroot
  fontconfig-config fonts-dejavu-core fonts-dejavu-mono fonts-mathjax g++ g++-13 g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base
  gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu idle idle-python3.12 javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libaom3
  libasan8 libatomic1 libbinutils libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0 libctf0 libde265-0 libdeflate0 libdpkg-perl libexpat1-dev
  libfakeroot libfile-fcntllock-perl libfontconfig1 libgcc-13-dev libgd3 libgomp1 libgprofng0 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265
  libheif1 libhwasan0 libisl23 libitm1 libjbig0 libjpeg-turbo8 libjpeg8 libjs-jquery libjs-mathjax libjs-sphinxdoc libjs-underscore liblerc4 liblsan0 libmpc3
  libpython3-dev libpython3.12-dev libpython3.12-testsuite libquadmath0 libsframe1 libsharpyuv0 libstdc++-13-dev libtiff6 libtk8.6 libtsan2 libubsan1 libwebp7 libxft2
```

Clone repository

git clone https://github.com/gowthamsaikadali/vcube-3tier-application-python.git

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@web-server:/home/gowthamazure# git clone https://github.com/gowthamsaikadali/3-tier-application-in-linux-vm.git
Cloning into '3-tier-application-in-linux-vm'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 18 (delta 1), reused 18 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 9.74 KiB | 9.74 MiB/s, done.
Resolving deltas: 100% (1/1), done.
root@web-server:/home/gowthamazure# ls
3-tier-application-in-linux-vm
```

3-Tier Application Deployment on Linux VMs

Navigate to frontend directory

cd /web_server

```
root@web-server:/home/gowthamazure# cd 3-tier-application-in-linux-vm/
root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm# ls
app_server  web_server
root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm# cd web_server/
root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# ls
app.py  requirements.txt  static  templates
```

Setup Virtual Environment

Create virtual environment

python3 -m venv venv

```
root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# python3 -m venv venv
root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# source venv/bin/activate
```

Activate it

source venv/bin/activate

```
root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# python3 -m venv venv
root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# source venv/bin/activate
(venv) root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# pip install -r requirements.txt
```

Install Dependencies

Install required libraries

pip install -r requirements.txt

```
(venv) root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# pip install -r requirements.txt
Collecting Flask==2.2.5 (from -r requirements.txt (line 1))
  Downloading Flask-2.2.5-py3-none-any.whl.metadata (3.9 kB)
Collecting requests==2.31.0 (from -r requirements.txt (line 2))
  Downloading requests-2.31.0-py3-none-any.whl.metadata (4.6 kB)
Collecting Werkzeug>=2.2.2 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading werkzeug-3.1.7-py3-none-any.whl.metadata (4.0 kB)
Collecting Jinja2>=3.0 (from Flask==2.2.5->-r requirements.txt (line 1))
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting itsdangerous>=2.0 (from Flask==2.2.5->-r requirements.txt (line 1))
```

Configure Environment

export APP_INTERNAL="http://<APP_PRIVATE_IP>:5001"

Connect frontend to backend

export WEB_HOST="0.0.0.0"
export WEB_PORT=5000

Set frontend host and port

```
(venv) root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# export APP_INTERNAL="http://10.0.1.4:5001"
export WEB_HOST="0.0.0.0"
export WEB_PORT=5000
```

Allow port and run frontend in background

nohup python3 app.py > web_server.log 2>&1 &

```
(venv) root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# nohup python3 app.py > web_server.log 2>&1 &
[1] 8489
(venv) root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server#
```

ACCESS THE APPLICATION

Open your browser and visit:

http://<PUBLIC_IP_OF_WEB_SERVER>:5000

The Web Server proxies API requests to:

http://<APP_SERVER_PRIVATE_IP>:5001

which connects to:

<DB_SERVER_PRIVATE_IP>:3306

STEP 5: VERIFY CONNECTIVITY

From App Server → DB Server

mysql -h <DB_SERVER_PRIVATE_IP> -u appuser -p

```
(venv) root@application-server:/home/gowthamazure/3-tier-application-in-linux-vm/app_server# mysql -h 10.0.0.4 -u appuser -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

From Web Server → App Server

curl http://<APP_SERVER_PRIVATE_IP>:5001/api/health

Expected:

{"status": "ok"}

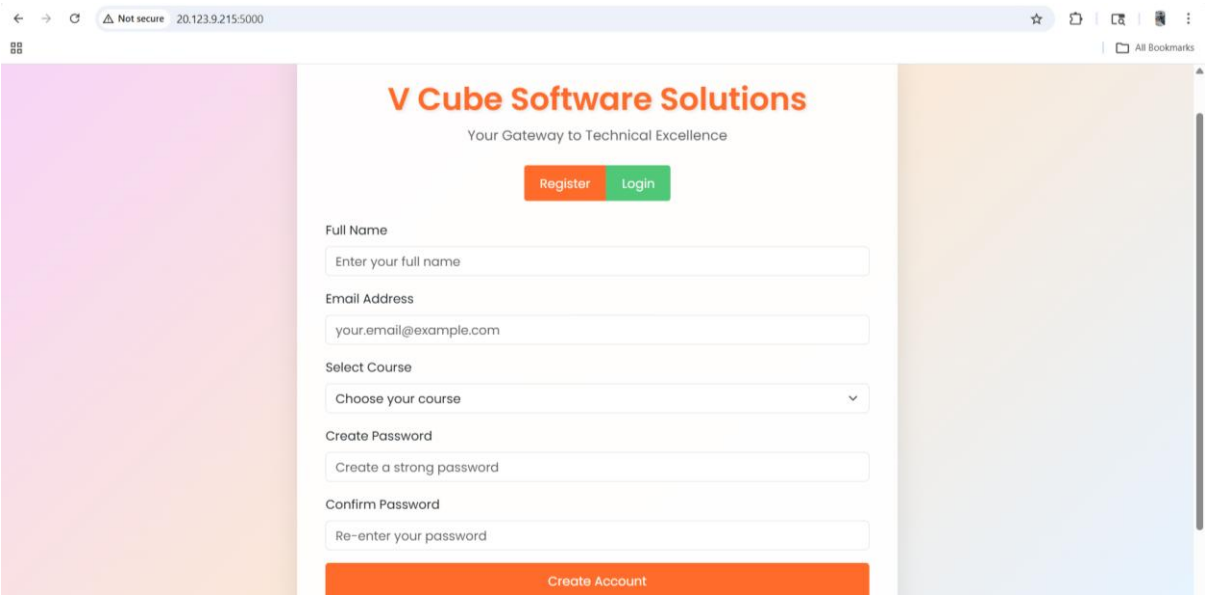
```
(venv) root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server# curl http://10.0.1.4:5001/api/health
{
  "status": "ok"
}
(venv) root@web-server:/home/gowthamazure/3-tier-application-in-linux-vm/web_server#
```

From Browser → Web Server

http://<PUBLIC_IP_WEB_SERVER>:5000

You should see the VCube App interface.

RESULT SCREENSHOT



TO VIEW DATABASE

3-Tier Application Deployment on Linux VMs

```
root@database-server:/home/gowthamazure# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 16
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE appdb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM user;
+----+----------------------------+--------------------+------------+--------------------------
```
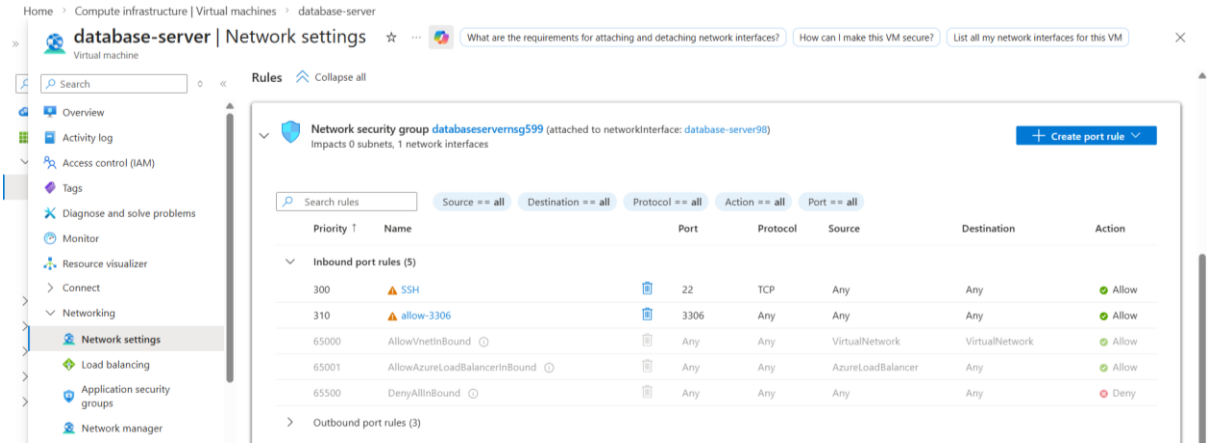

3-Tier Application Deployment on Linux VMs



Port 22 → SSH (for remote access to all servers)

These ports must be open for communication between respective servers.

SUMMARY OF IP FLOW

Component	Example Private/Public IP	Port	Description
Web Server	<public_ip_web>	5000	User interface (frontend)
App Server	<private_ip_app>	5001	Flask API + JWT + Logic
DB Server	<private_ip_db>	3306	MySQL database

Conclusion

This project demonstrates the successful deployment of a 3-tier application by separating the Web, App, and Database layers. It improves scalability, security, and maintainability.

Through this project, I gained hands-on experience in application deployment, Linux server setup, Flask–MySQL integration, and real-world troubleshooting.